Personalised Federated Learning for District Heating Demand Estimation under Portfolio Heterogeneity

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Abstract. District heating (DH) operators managing heterogeneous building portfolios face a structural conflict: accurate demand estimation requires building-level data, yet centralising that data is both privacy-sensitive and—as we show—technically counterproductive. We present a demand-driven federated learning (FL) framework in which buildings estimate their own heat energy consumption locally while a housing company aggregates per-building forecasts for portfolio planning. A learning channel exchanges only model weight updates (21.3 KB/round) via a streaming protocol; an operational channel delivers forecasts without raw data ever leaving the building.

We evaluate nine scenarios on 250 buildings (CoV = 1.33) from a 988-substation Swedish DH network. Centralised data pooling degrades accuracy by $35.4 \pm 3.0\%$ relative to local training ($p<0.001$, all five seeds), confirming that FL's privacy constraint sacrifices no accuracy. Personalised FL matches local training per-building (MAE 0.696 ± 0.014 vs. 0.682 ± 0.026 kWh/h, Wilcoxon non-significant in 4/5 seeds, with half the variance), shows a directional advantage at longer horizons, and outperforms it at portfolio level (MAPE 3.9%, 32% better than independent local models). Cross-dataset validation on Danish DH subsets shows FL's value correlates inversely with heterogeneity, providing a deployment guideline: FL is most justified where heterogeneity makes centralisation most harmful. The framework trains in 67 min (2.2× faster than local training) and exchanges 207.6 MB (5.6× less than raw-data centralisation).

Keywords: Federated learning · District heating · Demand estimation · Energy forecasting · Privacy-preserving machine learning

1 Introduction

District heating (DH) systems across Northern and Central Europe operate under a supply-driven paradigm in which individual building consumption remains opaque to housing companies, constraining demand-side optimisation and peak-load management [24, 9]. The natural remedy pooling substation data centrally is privacy sensitive and, as this paper demonstrates, technically counterproductive under the extreme heterogeneity (CoV>1.0) characteristic of mixed-use DH portfolios. We propose a demand-driven federated learning (FL) framework in which each building estimates its own heat consumption locally while a housing company aggregates per-building forecasts for portfolio planning. Buildings exchange only model weight updates (learning channel) and receive rolling demand forecasts (operational channel), with all raw measurements retained behind a privacy boundary aligned with GDPR data minimisation principles [15, 6].

While prior FL building energy forecasting relies on small, homogeneous populations (10–40 buildings, CoV < 0.5) where collaborative benefits are easily realized, this paper tackles the substantially harder setting of DH frameworks:

> *To what extent can a demand-driven, personalised federated learning framework provide accurate, scalable, and privacy-preserving estimation of heat energy consumption across heterogeneous DH buildings?*

The main contributions are:

1. A demand-driven FL framework with two channels—a learning channel ($\approx$ 21.3 KB/round) and an operational channel—using a streaming protocol where each communication round trains on new chronological data. A systematic nine-scenario comparison on 250 buildings (CoV = 1.33) establishing that centralised pooling is counterproductive ($35.4 \pm 3.0\%$ worse than local training, p<0.001 per seed), while personalised FL recovers per-building accuracy to within 2% of local training (non-significant in 4/5 seeds, half the variance) with a directional advantage at longer horizons.
2. Portfolio-level validation achieving MAPE 3.9 % (32 % better than local models) and rolling weekly MAPE 5.0 %, confirming fleet-level demand intelligence without centralised data access.
3. Cross-dataset validation on Danish Aalborg benchmark subsets, showing FL's value correlates inversely with portfolio heterogeneity.

2 Related Work

Machine learning techniques are widely applied to short-term DH heat load prediction. Wei et al. [23] examined support vector regression, XGBoost, and LSTM models, showing that outdoor temperature dominates predictive power and tree-based methods offer competitive accuracy at reduced cost. Song et al. [19] introduced a hybrid CNN–LSTM architecture combining spatial and temporal modelling. Hua et al. [10] analysed DH demand at building and city scales, emphasising nonlinear effects that restrict linear methods. A shared assumption is centralised data access—all substation measurements pooled into a single repository. In operational DH systems, however, data are distributed across substations with different stakeholders, raising governance, privacy, and scalability concerns [11].

Federated learning for energy forecasting. Federated learning (FL) enables collaborative model training across distributed clients without transferring raw data [15]. Li et al. [12] applied FL with secure aggregation to 13 office buildings, showing federated models match centralised accuracy with a 25.4 % CV-RMSE improvement when target buildings have limited data. Wang et al. [21] introduced an adaptive FL framework for 10 campus buildings with anomaly detection, reporting a 10 % error reduction. Sarmento et al. [17] integrated CNN–LSTM with FL on edge devices. These studies share three features: small client populations (10–40 buildings), homogeneous building types, and static train/test splits. Within the DH domain, Dai et al. [5] proposed multi-centre FL for 24-hour demand forecasting on 2,459 single-family houses, outperforming standard FL by 13.86% through cluster-based aggregation. However, the dataset is restricted to a single homogeneous building type. Badgujar [2] previously assessed FedAvg and FedAdam on three simulated DH substations with two input features. The present work extends both lines through the first DH-FL evaluation on a diverse portfolio at a scale of hundreds of substations.

Personalisation in heterogeneous FL. Statistical heterogeneity (non-IID client data) is the primary challenge in FL [14], and the personalised FL literature offers several response strategies: regularisation (FedProx [14], proximal term); adaptive server-side optimisation (FedOpt [16]: FedAdam, FedYogi, FedAdagrad); clustering (IFCA [8], one global model per gradient-similarity cluster); architectural splitting (FedPer [1], shared base with per-client head); meta-learning (PerFedAvg [7], fast local adaptation); local–global proximal coupling (Ditto [13], pFedMe [20]); and post-hoc local fine-tuning [25, 4] – the established and operationally simplest baseline. Most directly related, Wang et al. [22] proposed Metaformer, a transformer-based personalised FL evaluated on 40 buildings (gains 10–40%). The present study extends personalised FL to district heating at substantially larger scale (250 buildings, validated to 988) and far greater heterogeneity (CoV $= 1.33$ vs. typical < 0.5), using a streaming protocol and validating operational utility through portfolio-level forecasting. We position post-hoc fine-tuning relative to architectural alternatives in Sect. 4 and outline their comparative evaluation as follow-up work in Sect. 6.3.

Gap statement. No prior study evaluates FL for DH demand estimation against strong non-federated baselines (per-building XGBoost, centralised MLP), across forecast horizons from 1 hour to 7 days, at the scale of hundreds of real substations under extreme heterogeneity (CoV > 1), with a streaming FL protocol, and with cross-dataset validation. Furthermore, no existing study validates FL's operational utility through portfolio-level demand forecasting—the capability that makes FL valuable for housing companies. This paper addresses these gaps.

3 Problem Formulation

For building $k \in \{1, \ldots, K\}$ in a DH network, let $q_{k,t}$ denote heat energy consumption (kWh) at hour t and $\mathbf{x}_{k,t}$ the operational measurement vector (flow rate, supply/return temperatures, ΔT, outdoor temperature).

The local feature vector for building k at hour t is:

$$\mathbf{z}_t^{(k)} = \phi\Big(q_t^{(k)}, \ldots, q_{t-L}^{(k)},\ \mathbf{x}_t^{(k)}, \ldots, \mathbf{x}_{t-L}^{(k)},\ \boldsymbol{\tau}_t\Big) \tag{1}$$

where L is the lookback window (up to 168 hours), $\phi(\cdot)$ produces lagged values, rolling statistics, and rate-of-change features, and $\boldsymbol{\tau}_t$ encodes temporal information (hour-of-day, day-of-week, month) via sinusoidal encoding. The estimation task learns a mapping f_θ that predicts:

$$\hat{q}_{t+h}^{(k)} = f_\theta\Big(\mathbf{z}_t^{(k)}\Big) \tag{2}$$

where θ denotes the model parameters and $h \in \{1, 6, 24, 168\}$ hours.

Learning objectives. Table 1 defines nine estimation scenarios forming a progressive comparison chain across model family (XGBoost, MLP), data regime (local, centralised, federated), aggregation strategy (FedAvg [15], FedAdam [16], FedProx [14]), and personalisation via post-hoc fine-tuning (Algorithm 1, lines 12–14).

4 Demand-Driven Estimation Framework

The nine training paradigms are evaluated on the same dataset under identical settings, isolating the effect of training strategy on estimation quality. The demand-driven FL framework operates through two information flows separated by a privacy boundary, illustrated in Fig. 1.

4.1 Prediction Models and FL Strategies

Table 1 provides an overview of the nine estimation scenarios. The XGBoost scenarios (1–2) provide a strong gradient-boosted baseline from a different model family; the MLP scenarios (3–9) share an identical architecture so that differences isolate the effect of training strategy.

Model architecture. All MLP scenarios employ a two-hidden-layer architecture with [64, 32] units, ReLU activations, and the Adam optimiser, yielding 5,441 trainable parameters. XGBoost uses 200 estimators with max depth 6.

FL aggregation strategies. We evaluate three server-side strategies: FedAvg [15] (weighted averaging), FedAdam [16] (adaptive server-side optimisation), and FedProx [14] (proximal regularisation $\mu/2\|\theta - \theta^{(r)}\|^2$).

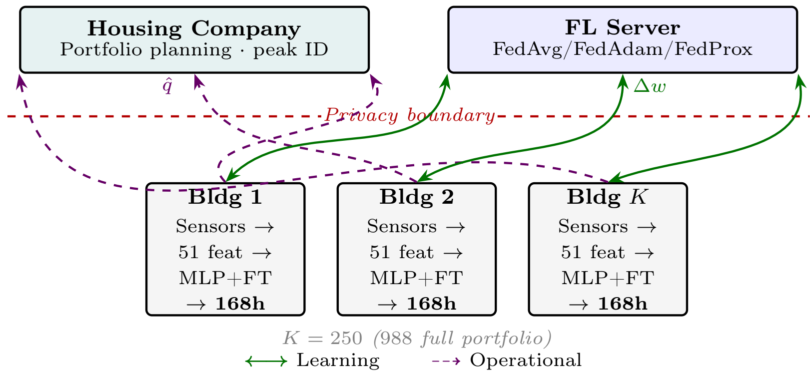

Fig. 1. Demand-driven FL framework. Learning channel (green) exchanges weights (≈21.3 KB/round). Operational channel (violet) delivers 168h forecasts directly to housing company. Raw data never leave buildings. Horizons $h \in \{1, 6, 24, 168\}$; 168h supports weekly procurement (Section 5.3).

Table 1. Overview of the nine estimation scenarios.

#	Scenario	Model	Training	Data shared
1	Local XGBoost	XGBoost	Per-building	None
2	Centralised XGBoost	XGBoost	All pooled	All raw data
3	Local MLP	MLP	Per-building	None
4	FedAvg	MLP	FL (FedAvg)	Weights only
5	FedAdam	MLP	FL (FedAdam)	Weights only
6	FedProx	MLP	FL (FedProx)	Weights only
7	Pers. FL (FedAdam)	MLP	FL + fine-tune	Weights only
8	Pers. FL (FedProx)	MLP	FL + fine-tune	Weights only
9	Centralised MLP	MLP	All pooled	All raw data

Streaming FL protocol and personalisation. Algorithm 1 summarises the complete training procedure. Unlike conventional FL where all rounds train on a fixed dataset, each round uses a chronologically distinct segment, emulating continuous deployment. After FL converges, each building fine-tunes the global model on a held-out adaptation set (15% of data), never accessed during FL training. We adopt post-hoc fine-tuning as the established baseline (Sect. 2).

4.2 Experimental Setup

Dataset. The dataset consists of hourly measurements collected from 988 district heating substations operated by a municipal housing company in a mid-sized Swedish municipality. The observation period spans January 2019 to September 2024 and includes approximately 37.4 million hourly records. For each substation, the available variables are energy consumption (kWh/h), volumetric flow rate (m^3/h), supply temperature, return temperature, and delta temperature.

Algorithm 1 Streaming personalised FL for a DH portfolio

Initialise global parameters $\theta^{(0)}$
Partition each building's training data into R chronological segments
for round $r = 1$ to R **do**
  Server broadcasts $\theta^{(r-1)}$ to all K buildings
  for each building $k = 1, \ldots, K$ **in parallel do**
    Train from $\theta^{(r-1)}$ on segment $D_k^{(r)}$ for E local epochs $\rightarrow \theta_k^{(r)}$
    Send $\theta_k^{(r)}$ to server
  end for
  Aggregate $\{\theta_k^{(r)}\}$ via FedAvg / FedAdam / FedProx $\rightarrow \theta^{(r)}$
end for
$\theta^*_{\text{global}} \leftarrow \theta^{(R)}$
for each building $k = 1, \ldots, K$ **do**
  Fine-tune from θ^*_{global} on held-out $D_k^{\text{adapt}} \rightarrow \theta^*_k$
end for
return $\{\theta^*_k\}_{k=1}^{K}$ {building-specific personalised models}

Data cleaning. One substation with 100% negative delta-temperature readings (disconnected meter) was removed. For the remaining 988 buildings, negative delta-temperature values (0.5% of observations, mainly summer standby periods) were clipped to zero. Outdoor temperature data were sourced from SMHI using the nearest weather station.

Feature engineering. The input feature vector $\mathbf{z}_t$ (Eq. 1) comprises 51 engineered features: lagged consumption values ($y_{t-1}, \ldots, y_{t-24}, y_{t-168}$); rolling statistics (mean, max, std over 6/12/24 h windows); sine–cosine temporal encodings of hour, day-of-week, and month; and nine outdoor temperature features. Input features are z-score normalised per building; the target variable y (kWh/h) is not normalised.

Data heterogeneity. The coefficient of variation (CoV) of mean annual energy consumption across buildings is 1.33, reflecting extreme heterogeneity. For comparison, the Aalborg benchmark (Section 5.4) exhibits CoV = 0.41–0.46.

Hyperparameters. MLP scenarios use Adam at learning rate 0.001 with learning rate decay $\eta/(1 + 10^{-5}r)$ (where r is the round index) and per-building batch size 64; the centralised MLP (Scenario 9) uses batch size 1024 over 5.68M pooled rows. FL strategies run for $R = 20$ rounds with full client participation and 10 local epochs per round; FedProx sets $\mu = 0.01$. The learning channel exchanges ≈21.3 KB of weights per client per round; the operational channel delivers ≈1.3 KB forecasts per building.

Fine-tuning implementation. Post-hoc fine-tuning continues training the full global FL model θ^*_{global} on each building's adaptation partition D_k^{adapt} — the chronologically late 15% of that building's data, disjoint from the held-out 15%

test partition and unseen during FL training. All layers are fine-tuned with the same mean-squared-error loss as FL training. The Adam optimiser is re-initialised (no momentum carryover) with $\eta_{ft} = 10^{-3}$, batch size 64, up to 10 epochs, and early stopping (patience 2) on the building's Val 10% partition. Per-building Adapt sizes range ∼5,000–8,000 hourly samples. Fine-tuning operates entirely on local data; no information crosses the privacy boundary.

Multi-seed evaluation. For the four primary MLP scenarios (Local MLP, Centralised MLP, Personalised FL with FedAdam and FedProx bases) and FedAdam global, we run five independent trials with seeds $\{42, 43, 44, 45, 46\}$; results are reported as mean±std. Each seed independently controls Python `random`, NumPy, TensorFlow, and `PYTHONHASHSEED`. FedAvg, FedProx, and XGBoost baselines are reported on seed=42, since FedAdam global (std= 0.016) confirms the global FL family is seed-stable.

Hardware and software. Implementation uses the Flower FL framework [3] with a TensorFlow backend, simulated on an NVIDIA DGX Station (4× V100-32GB GPUs). All 250 building clients participate in every communication round.

Data partitioning. For each building, data are split chronologically into four disjoint partitions:

Train (60%)	Adapt (15%)	Val (10%)	Test (15%)
FL rounds	Fine-tuning	Early stopping	Evaluation

XGBoost scenarios use a 70/15/15 split (no adaptation phase); the test set begins at the same 85% temporal mark across all scenarios.

Building selection and outlier handling. A deterministic subset of 250 buildings from the full 988 is used across all scenarios. Evaluation metrics are reported for 246 buildings after excluding four with train-to-test consumption ratio exceeding 1.3× (regime changes). Including all 250 does not alter relative rankings but inflates Local MLP mean MAE from 0.682 to 0.958 due to one extreme outlier (MAE = 64.28 kWh/h from a regime change, vs. 3.76 kWh/h for Personalised FL on the same building).

Evaluation metrics. Mean Absolute Error (MAE, kWh/h) is the primary metric, reported as the mean across buildings. Median MAE accounts for the heavy-tailed distribution. The coefficient of determination (R^2) quantifies explained variance. Pairwise significance is assessed using Wilcoxon signed-rank tests ($\alpha = 0.05$).

Portfolio-level and cross-dataset evaluation. Portfolio-level forecasts aggregate hourly predictions across 247 buildings (3 excluded for $< 2{,}000$ test hours). Cross-dataset validation uses the Aalborg smart heat meter dataset [18] (3,021 Danish buildings): a 50-building single-family subset (CoV = 0.41) and a 100-building mixed-type subset (CoV = 0.46).

Table 2. Per-building estimation accuracy (h = 1, 246 buildings). MAE in kWh/h. Multi-seed scenarios as mean±std across 5 seeds (42–46); † marks single-seed (seed=42). Best MLP result per column in bold.

#	Scenario	MAE	Median	R^2 mean	R^2 med
2	Centralised XGB	0.590±0.001	0.510	0.465±0.0001	0.465
3	Local MLP	**0.682±0.026**	0.582	0.43±0.04	0.452
7	Pers. FL (FedAdam)	0.696±0.014	**0.575**	0.41±0.02	**0.450**
8	Pers. FL (FedProx)	0.698±0.014	0.605	**0.45±0.004**	0.453
1	Local XGBoost	0.798±0.003	0.674	0.38±0.004	0.406
9	Centralised MLP	1.054±0.024	0.985	0.19±0.02	0.273
6	FedProx†	1.133	0.984	0.209	0.285
4	FedAvg†	1.146	0.983	0.205	0.286
5	FedAdam	1.142±0.016	0.906	0.21±0.01	0.276

5 Results

Reporting conventions. Multi-seed results (mean±std across seeds $\{42, \ldots, 46\}$) appear in Tables 2 and 3 and Sec. 5.1; all other results in this section, including multi-horizon, portfolio, cross-dataset, scalability, and training-time measurements, are reported on $seed = 42$.

5.1 Per-Building Estimation Accuracy

Table 2 summarises per-building estimation accuracy for all nine scenarios at $h = 1$. Centralised XGBoost achieves the lowest MAE at 0.590 kWh/h, an upper bound from a richer model family. Among MLP-based scenarios, Local MLP achieves the lowest mean MAE (0.682 kWh/h).

Table 2 yields three findings: (i) personalisation closes 65 of the 67% gap between unpersonalised FL and local training (FedAdam global 1.142 ± 0.016 vs. Personalised FL 0.696 ± 0.014 vs. Local MLP 0.682 ± 0.026); (ii) centralised pooling is consistently counterproductive (35.4±3.0% worse, per-seed range 30.5–38.4%, $p<0.001$); and (iii) aggregation strategy choice (FedAvg/FedAdam/FedProx) matters far less than personalisation (global spread 0.013 kWh/h).

Stability and sensitivity across seeds. Personalised FL is substantially more stable than Local MLP (std 0.014 vs. 0.026 kWh/h), with Local MLP variance driven by a single substation (B045) whose limited data causes MAE to range 1.86–17.65 kWh/h across seeds. A sensitivity analysis excluding B045 (n = 245) inverts the apparent FedAdam-vs-Local ordering: Personalised FedAdam outperforms Local MLP in all five seeds (mean $\Delta = -1.5\%$; per-seed range $[-3.5\%, -0.2\%]$). This supports the selective-deployment recommendation (Sect. 6.2 and Fig. 2): when a single substation's training instability can determine the portfolio-level winner, building-specific deployment decisions matter more than blanket strategy selection.

Table 3. Per-seed Wilcoxon signed-rank tests ($h = 1$, 246 buildings, $\alpha = 0.05$). Paired per-building MAE differences across 5 seeds (42–46). ΔMAE $= (\overline{MAE}_A - \overline{MAE}_B)/\overline{MAE}_B$; negative favours method A. Significant seeds = number of seeds with $p < 0.001$.

Comparison	ΔMAE range	median p	Sig. seeds
Pers. FedAdam vs Local MLP	$[-3.7, +6.0]\%$	1.6×10^{-1}	1/5
Pers. FedProx vs Local MLP	$[-4.6, +6.2]\%$	6.6×10^{-10}	4/5 ***
Pers. FedAdam vs Centr. MLP	$[-35.2, -32.2]\%$	1.2×10^{-37}	5/5 ***
Local MLP vs Centr. MLP	$[-38.4, -30.6]\%$	8.2×10^{-38}	5/5 ***
Pers. FedAdam vs Pers. FedProx	$[-1.3, +1.0]\%$	6.9×10^{-16}	5/5 ***

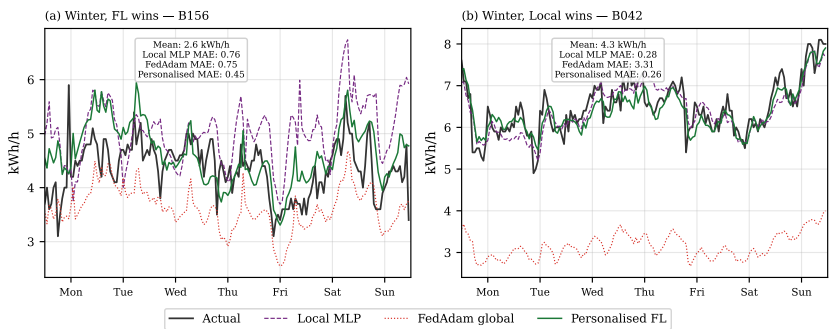

Fig. 2. Per-building prediction comparison ($h = 1$, first week of test set). (a) Building where Personalised FL wins: cross-building knowledge transfer recovers patterns better than local data alone. (b) Building where Local MLP wins: local training suffices and FL's shared representation introduces a small bias. This complementary behaviour motivates selective deployment (per-building oracle MAE = 0.637; FL wins on 50 % of buildings).

Personalised FL's median MAE (0.575) is lower than Local MLP's (0.582), confirming FL wins for the majority of buildings; the higher mean reflects a small minority of buildings where fine-tuning under-adapts (Fig. 2). Figure 2 shows this complementarity for two representative buildings: one where Personalised FL outperforms Local MLP, and one where Local MLP wins.

5.2 Multi-Horizon Analysis

Multi-horizon evaluation extends the per-building $h = 1$ results of Table 2 to $h \in \{6, 24, 168\}$ hours (Table 4). For direct comparability across the three longer horizons, all values in Table 4 are reported on seed=42. At $h = 24$, Personalised FL (FedAdam) reaches MAE 0.832 versus Local MLP's 0.844; at $h = 168$, Personalised FL (FedProx) reaches 1.027 versus Local MLP's 1.038. These per-horizon differences (1.1–1.4%, seed=42) are smaller than the Local MLP across-seed standard deviation observed at $h = 1$ (0.026, Table 2); a multi-seed multi-horizon evaluation would be required to establish statistical separation and is

Table 4. Multi-horizon estimation accuracy (MAE, kWh/h, 246 buildings, $seed = 42$). Best MLP result per horizon in bold. For $h = 1$ see Table 2.

Scenario	$h = 6$	$h = 24$	$h = 168$
Centralised XGB	0.698	0.775	0.955
Local MLP	**0.749**	0.844	1.038
Pers. FL (FedAdam)	0.774	**0.832**	1.056
Pers. FL (FedProx)	0.783	0.836	**1.027**
Local XGBoost	0.854	0.934	1.120
Centralised MLP	1.090	1.138	1.319
FedProx	1.207	1.226	1.340
FedAdam	1.194	1.238	1.338
FedAvg	1.224	1.241	1.357

Table 5. Portfolio-level demand forecast accuracy (247 buildings, 2,297 hours, Feb–May 2024).

Scenario	MAE (kWh/h)	MAPE (%)	R^2
Pers. FL (FedAdam)	**35.2**	**3.9**	**0.612**
Local MLP	48.1	5.7	0.328
Centralised MLP	48.6	5.7	0.439
FedAdam Global	212.1	24.2	−7.092

left for future work. FedProx-based personalisation outperforms FedAdam at $h = 168$ (1.027 vs. 1.056), suggesting the proximal term limits parameter drift at longer horizons.

5.3 Portfolio-Level Demand Forecasting

To evaluate the operational channel, per-building predictions are aggregated hourly across 247 buildings over 95 days (Table 5). Portfolio-level metrics in this section are reported on seed=42; per-building multi-seed sensitivity for the same scenarios is reported in Table 2.

Personalised FL achieves MAPE 3.9 %, a 32% reduction over Local MLP and Centralised MLP (each 5.7%). A notable observation: at the per-building level Centralised MLP is $35.4 \pm 3.0\%$ worse than Local MLP (Sect. 5.1), yet at portfolio level their MAEs converge (48.6 vs. 48.1), reflecting partial cancellation of opposite-signed per-building errors. Personalised FL achieves lower portfolio MAE (35.2) (Fig. 3), by producing residuals that are both smaller and less correlated, enabling more effective cancellation. The portfolio R^2 of 0.612 (vs. 0.328 for Local MLP) confirms FL captures substantially more variance in aggregate demand.

A rolling 168-hour forecast simulation over 53 days (250 buildings, Sep–Oct 2023) further confirms operational viability: Personalised FL achieves weekly demand MAPE 5.0 % and daily MAPE 6.6 % (Fig. 4), supporting the housing company's procurement cycle.

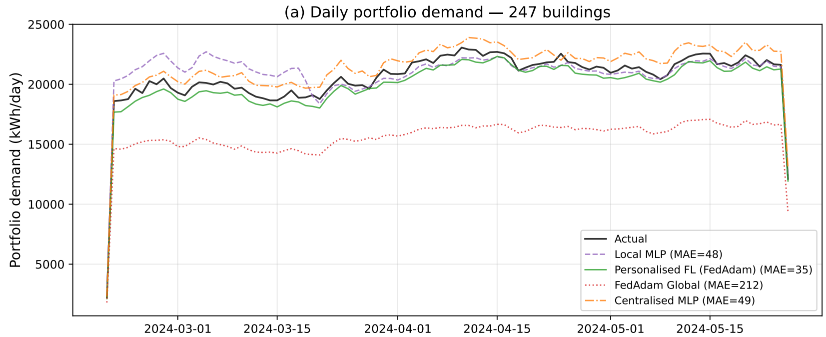

Fig. 3. Portfolio-level demand forecast (247 buildings, Feb–May 2024). Pers. FL (MAE 35), Local MLP (48), Centralised MLP (49), FedAdam Global (212).

Table 6. Cross-dataset validation: FL performance gap as a function of portfolio heterogeneity (CoV). Mean MAE across buildings in kWh/h; all rows report seed=42 single-seed for direct cross-dataset comparability.

Dataset	K	CoV	Local	Pers. FL	Gap
Swedish DH network	250	1.33	0.669	0.690	+3.2%
Danish DH (SF)	50	0.41	0.495	0.611	+23.6%
Danish DH (mixed)	100	0.46	0.485	0.577	+19.0%

5.4 Cross-Dataset Validation

To assess generalisability, the full pipeline is applied to two subsets of the Danish DH smart heat meter dataset [18] (Table 6).

The gap decreases monotonically with heterogeneity: from +23.6% at CoV = 0.41 to +3.2% at CoV = 1.33. In homogeneous settings, local models suffice and FL aggregation adds noise. Under extreme heterogeneity, personalisation recovers building-specific performance while maintaining data locality.

5.5 Scalability and Computational Efficiency

Scalability experiments across $K \in \{20, 50, 100, 200, 250, 500, 988\}$ clients confirm that personalised FL maintains stable accuracy (MAE 0.67–1.18) while global FL variants are volatile (MAE 1.11–1.98). At $K = 988$ (full network), Personalised FL achieves MAE 0.751, within 0.09 kWh/h of the $K = 50$ result, demonstrating the framework scales to network-wide deployment. Personalisation consistently improves over global FL by 40–55 % at every scale.

Personalised FL trains in 67 minutes (2.2× faster than Local MLP at 146 min), exchanging 21.3 KB per client per round (207.6 MB total)—a 5.6× reduction compared to centralising raw data (1,159 MB). The Centralised MLP requires the longest training time (156 min) while delivering the weakest MLP perfor-

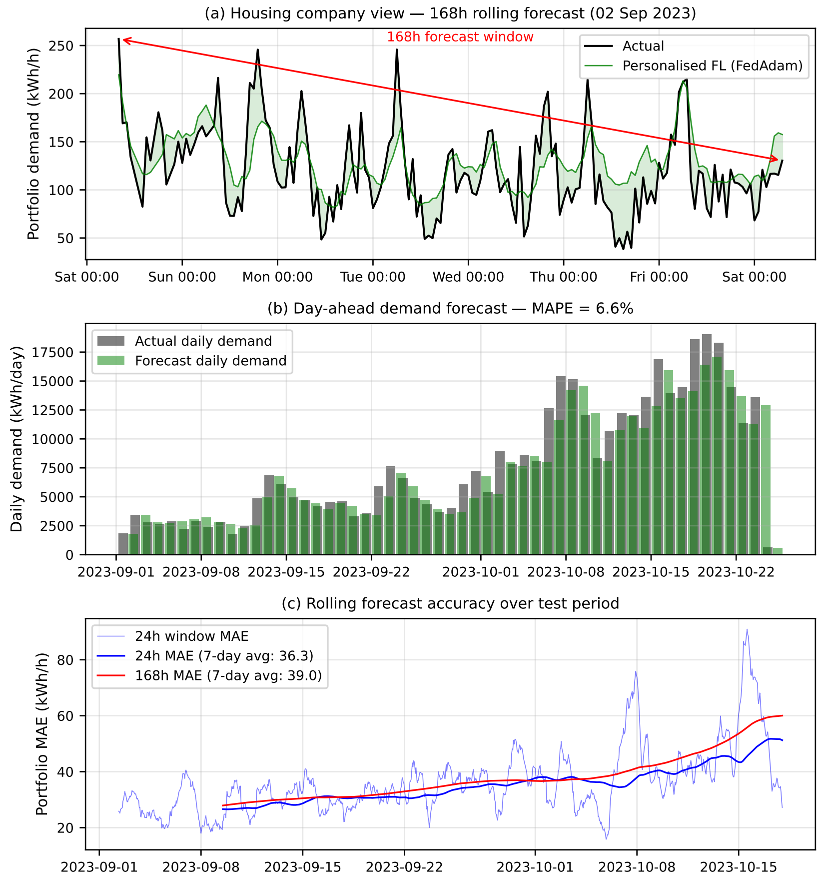

Fig. 4. Rolling 168-hour demand forecast simulation (250 buildings, 53 days). Weekly demand MAPE = 5.0 %, daily MAPE = 6.6 %.

mance. FL is thus not only privacy-preserving but also computationally and communication efficient.

6 Discussion

6.1 Centralised Pooling Fails Under Heterogeneity

The $35.4 \pm 3.0\%$ performance degradation of Centralised MLP relative to Local MLP (per-seed range 30.5% to 38.4%, $p < 0.001$ at each seed) constitutes the most decisive result. Under extreme heterogeneity (CoV = 1.33), pooling data from buildings with mean demands ranging from 0.3 to 28.1 kWh/h obscures building-specific dynamics. This extends prior evidence that non-IID distributions undermine centralised learning to the DH domain, where heterogeneity is structural—driven by differences in building size, construction quality, occupancy, and heating systems.

6.2 FL's Value: Portfolio Intelligence, Not Per-Building Accuracy

At the per-building level, Personalised FL achieves accuracy statistically comparable to Local MLP (non-significant in 4/5 seeds; mean MAE within $\sim 2\%$; see Table 3). Under extreme heterogeneity, the global FL model serves as an informed initialisation rather than a complete predictor, with building-specific patterns recovered during fine-tuning. FL thus functions as a privacy-preserving mechanism for sharing representation-level information. Nevertheless, FL exhibits increased robustness: the most extreme outlier building shows Local MLP MAE of 64.28 kWh/h following a regime change, versus 3.76 kWh/h for Personalised FL.

The principal value emerges at portfolio level. Aggregating Personalised FL predictions yields fleet-level MAPE 3.9 % (32% better than local models), because FL produces less correlated prediction errors: the shared representation acts as an implicit regulariser, enabling more effective cancellation under aggregation. The rolling 168-hour MAPE of 5.0 % meets procurement planning requirements while preserving data locality. Per-building oracle analysis (MAE 0.637, FL wins on 50% of buildings; Fig. 2 illustrates two complementary cases) further motivates selective deployment.

Cross-dataset validation shows FL's value is governed by heterogeneity: FL degrades by 23.6% at $\mathrm{CoV} = 0.41$ but only 3.2% at $\mathrm{CoV} = 1.33$ after personalisation. This yields a deployment guideline: FL is most justified for highly heterogeneous portfolios.

6.3 Positioning, Limitations, and Future Work

Several limitations apply. Personalisation in this study uses post-hoc local fine tuning [25, 4]; architectural personalisation methods such as FedPer [1], Ditto [13], and meta-learning-based approaches [7] represent natural extensions whose comparative performance under the heterogeneity regime examined here ($\mathrm{CoV} = 1.33$) remains an open question. Three further questions remain open. The streaming protocol assumes full client participation per round; in deployed systems, dropout or stragglers would require partial-participation handling. Neither adversarial robustness (poisoning, membership inference) nor differential privacy guarantees were evaluated. Per-building z-score normalisation allows the same input value to carry different absolute meaning across clients, shifting recovery of building-specific scale onto fine-tuning; the Centralised MLP's 35.4% degradation indicates the shared representation alone cannot resolve scale heterogeneity. Other limitations include the MLP architecture choice (favouring communication efficiency over sequence-model accuracy), and the absence of live-data field deployment. Additional future work includes incorporating building size, construction year, occupancy patterns and the integration of differential privacy guarantees over the FL learning channel.

7 Conclusion

This paper investigated whether a demand-driven, personalised FL framework can provide accurate, scalable, and privacy-preserving estimation of heat energy consumption across heterogeneous DH buildings. Through a controlled comparison of nine scenarios on 250 buildings ($\text{CoV} = 1.33$), the answer is affirmative — with a reframing of where FL's value lies. The controlled nine-scenario comparison confirms that FL's privacy constraint does not sacrifice accuracy: centralised pooling is counterproductive under structural heterogeneity, while personalised FL matches local training per-building and delivers a 32% portfolio-level improvement unreachable by any local-only approach. Cross-dataset validation provides a deployable guideline: FL's value is greatest where heterogeneity is highest. For DH operators, the framework offers a deployable path to privacy-preserving demand intelligence, with training completing in 67 minutes (207.6 MB, 2.2× faster than local training). Future work includes building metadata integration, differential privacy guarantees, and field deployment with live substation data.

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References

1. Arivazhagan, M.G., Aggarwal, V., Singh, A.K., Choudhary, S.: Federated learning with personalization layers. arXiv preprint arXiv:1912.00818 (2019)
2. Badgujar, V.: Hourly Heat Load Prediction for District Heating Substations Using Federated Learning with Flower Framework: An Evaluation Study. Master's thesis, University of Borås (2025)
3. Beutel, D.J., Topal, T., Mathur, A., Qiu, X., Fernandez-Marques, J., Gao, Y., et al.: Flower: A friendly federated learning research framework. arXiv preprint arXiv:2007.14390 (2020)
4. Cheng, G., Chadha, K.N., Duchi, J.C.: Fine-tuning is fine in federated learning. arXiv preprint arXiv:2108.07313 (2021)
5. Dai, K., Fabello González, E., García-Betances, R.I.: A privacy preserving multicenter federated learning framework for district heating forecast. Energy and Buildings **328**, 115164 (2025)
6. European Parliament and Council of the European Union: Regulation (EU) 2016/679 (general data protection regulation). Tech. rep., Official Journal of the European Union (2016)
7. Fallah, A., Mokhtari, A., Ozdaglar, A.: Personalized federated learning with theoretical guarantees: A model-agnostic meta-learning approach. In: Advances in Neural Information Processing Systems (NeurIPS). vol. 33, pp. 3557–3568 (2020)

8. Ghosh, A., Chung, J., Yin, D., Ramchandran, K.: An efficient framework for clustered federated learning. In: Advances in Neural Information Processing Systems (NeurIPS). vol. 33 (2020)
9. Guelpa, E., Verda, V.: Demand response and other demand side management techniques for district heating: A review. Energy **219**, 119440 (2021). https://doi.org/10.1016/j.energy.2020.119440
10. Hua, P., Wang, H., Xie, Z., Lahdelma, R.: District heating load patterns and short-term forecasting for buildings and city level. Energy **289**, 129866 (2024). https://doi.org/10.1016/j.energy.2023.129866
11. Kairouz, P., McMahan, H.B., et al.: Advances and open problems in federated learning. Foundations and Trends in Machine Learning **14**(1–2), 1–210 (2021). https://doi.org/10.1561/2200000083
12. Li, J., Zhang, C., Zhao, Y., Qiu, W., Chen, Q., Zhang, X.: Federated learning-based short-term building energy consumption prediction method for solving the data silos problem. Building Simulation **15**(6), 1145–1159 (2022)
13. Li, T., Hu, S., Beirami, A., Smith, V.: Ditto: Fair and robust federated learning through personalization. In: International Conference on Machine Learning (ICML). pp. 6357–6368. PMLR (2021)
14. Li, T., Sahu, A.K., Zaheer, M., Sanjabi, M., Talwalkar, A., Smith, V.: Federated optimization in heterogeneous networks. Proceedings of Machine Learning and Systems (MLSys) **2**, 429–450 (2020)
15. McMahan, H.B., Moore, E., Ramage, D., Hampson, S., y Arcas, B.A.: Communication-efficient learning of deep networks from decentralized data. In: Artificial Intelligence and Statistics (AISTATS). pp. 1273–1282. Pmlr (2017)
16. Reddi, S., Charles, Z., Zaheer, M., Garrett, Z., Rush, K., Konečný, J., et al.: Adaptive federated optimization. arXiv preprint arXiv:2003.00295 (2020)
17. Sarmento, H., et al.: Forecasting energy power consumption using federated learning in edge computing devices. Internet of Things (2024)
18. Schaffer, M., Tvedebrink, T., Marszal-Pomianowska, A.: Three years of hourly data from 3021 smart heat meters installed in Danish residential buildings. Scientific Data **9**(1), 420 (2022)
19. Song, J., et al.: Predicting hourly heating load in a district heating system based on a hybrid CNN-LSTM model. Energy and Buildings **243**, 110998 (2021). https://doi.org/10.1016/j.enbuild.2021.110998
20. T. Dinh, C., Tran, N., Nguyen, J.: Personalized federated learning with moreau envelopes. In: Advances in Neural Information Processing Systems (NeurIPS). vol. 33, pp. 21394–21405 (2020)
21. Wang, R., Yun, H., Rayhana, R., Bin, J., Zhang, C., Herrera, O., et al.: An adaptive federated learning system for community building energy load forecasting and anomaly prediction. Energy and Buildings **295**, 113215 (2023)
22. Wang, R., Bai, L., Rayhana, R., Liu, Z.: Personalized federated learning for buildings energy consumption forecasting. Energy and Buildings **323** (2024). https://doi.org/10.1016/j.enbuild.2024.114762
23. Wei, Z., et al.: Prediction of residential district heating load based on machine learning: A case study. Energy **231**, 120661 (2021). https://doi.org/10.1016/j.energy.2021.120661
24. Werner, S.: District heating and cooling in Sweden. Energy **126**, 419–429 (2017). https://doi.org/10.1016/j.energy.2017.03.052
25. Yu, T., Bagdasaryan, E., Shmatikov, V.: Salvaging federated learning by local adaptation. arXiv preprint arXiv:2002.04758 (2020)